

Binafsha Bakhramova

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[LinkedIn](#) • [GitHub](#)

EDUCATION

Northeastern University, D'Amore-McKim School of Business Boston, MA
BS in Business Administration and Political Science, Minor in Data Science Expected May 2029
Concentrations: Business Analytics, FinTech, Public Policy **GPA: 3.7**

- **Honors & Activities:** Johns Martinson Honors Program, Dean's List, Women in Business; featured guest on NBC Townhouse discussing AI technology (June 2024)
- **Relevant Coursework:** Foundations of Data Science, Business Statistics, Financial Accounting for Advisors and Managers, International Relations

RESEARCH EXPERIENCE

Computational Redistricting Research | [Live Demo](#) | [GitHub](#) Boston, MA
Independent Research 2026 – Present

- Built a Markov Chain Monte Carlo simulation engine that generates ensembles of 500+ valid districting plans to test whether an observed map is a statistical outlier against its full distribution.
- Identified Texas's 2025 enacted map as an extreme outlier at the 0th percentile of the ensemble on fairness metrics, quantifying how far the observed case sits in the tail.

Northeastern University — Data Science Research Project | [Live Site](#) Leuven, Belgium
Machine Learning Developer May 2026 – Present

- Built and validated employment-forecasting models on a 2,354-row EU panel (27 countries, 8 sectors, 2012–2023), engineering an education-to-industry crosswalk and lag and change features from 2 Eurostat datasets.
- Caught that a near-perfect model was driven by look-ahead bias from a random data split, and corrected it with a temporal train/test design (train 2012–2020, test 2021–2023) to produce an honest out-of-sample estimate.
- Productionized the models into a Flask service backed by MySQL and built a Plotly dashboard for scenario inspection across countries and sectors.
- Built a sector reallocation model converting employment trends and graduate absorption into standardized demand scores and allocation shares, surfacing Computer Science as underfunded (+\$595K) and Engineering as overfunded (–\$655K) for Belgium.

Boston Area Research Initiative (BARI) Boston, MA
Research Assistant January 2026 – July 2026

- Built a real-time public heat-alert map for the Blue Hill Ave corridor (JavaScript, Open-Meteo API, Leaflet.js) turning live sensor readings into accessible visuals, contributing to data-pipeline and tool design for the Common SENSES sensing project.

NYU Tandon School of Engineering New York, NY
Research Assistant June 2024 – August 2024

- Benchmarked 3 control strategies for precision-robotics tracking by simulating NYU-Finger robot dynamics in PyBullet, then presented the most accurate, energy-efficient controller in a thesis-style defense.

TECHNICAL SKILLS

Languages: Python, R, SQL, JavaScript
Quantitative & Data: statistical modeling, regression, Monte Carlo simulation, model validation, time-series, Pandas, NumPy, scikit-learn
Tools & Frameworks: Excel, Flask, Streamlit, MySQL, Git/GitHub, Plotly, REST APIs
Languages (Spoken): English (Native), Russian (Professional Working Proficiency)

LEADERSHIP & ACTIVITIES

Northeastern University AI Advisory Board Boston, MA
Member April 2026 – Present

- Advising university leadership on institutional AI strategy across 3 focus areas, partnering with academic centers to expand student AI access and redesign classroom integration across Northeastern's 9 colleges.

EVH Hall Council Boston, MA
Vice President (Budget) February 2026 – April 2026

- Tracked and reported a \$7,500 budget (13% of EVH funding), monitoring spending against plan and preparing financial documentation for review and approval.

Accelerate Pre-College Programs Boston, MA
Student Life Ambassador June 2026 – Present

- Mentored 100+ first-time attendees in a 600-student pre-college program and helped run its community events.